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# Image Formation Model Demonstration
# Improving Image Quality for Autonomous Driving
# Google Colab
# =====

import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

# Upload image
uploaded = files.upload()

image_name = list(uploaded.keys())[0]

# Read image
img = cv2.imread(image_name)
img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

# -----
# Simulate Poor Image Formation
# (Blur + Low Brightness + Noise)
# -----

# Blur
blur = cv2.GaussianBlur(img, (11,11), 5)

# Reduce brightness
dark = np.clip(blur * 0.5, 0, 255).astype(np.uint8)

# Add Gaussian Noise
noise = np.random.normal(0, 20, dark.shape).astype(np.int16)
noisy = np.clip(dark.astype(np.int16) + noise, 0, 255).astype(np.uint8)

# -----
# Image Enhancement
# -----

# Brightness Enhancement
enhanced = cv2.convertScaleAbs(noisy, alpha=1.5, beta=30)

# Noise Reduction
enhanced = cv2.GaussianBlur(enhanced, (5,5), 0)

# Contrast Enhancement
lab = cv2.cvtColor(enhanced, cv2.COLOR_RGB2LAB)
l, a, b = cv2.split(lab)
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clahe = cv2.createCLAHE(clipLimit=2.0, tileGridSize=(8,8))
l = clahe.apply(l)

lab = cv2.merge((l,a,b))
enhanced = cv2.cvtColor(lab, cv2.COLOR_LAB2RGB)

# -----
# Display Results
# -----

plt.figure(figsize=(14,8))

plt.subplot(1,2,1)
plt.imshow(noisy)
plt.title("Poor Image (Blur + Noise + Low Light)")
plt.axis("off")

plt.subplot(1,2,2)
plt.imshow(enhanced)
plt.title("Enhanced Image")
plt.axis("off")

plt.tight_layout()
plt.show()

# -----
# Explanation
# -----

print("\nImage Formation Model Analysis")
print("-----")
print("Original image is degraded due to:")
print("- Blur")
print("- Low illumination")
print("- Sensor noise")

print("\nEnhancement Techniques Used:")
print("- Brightness correction")
print("- Noise reduction")
print("- Contrast enhancement (CLAHE)")

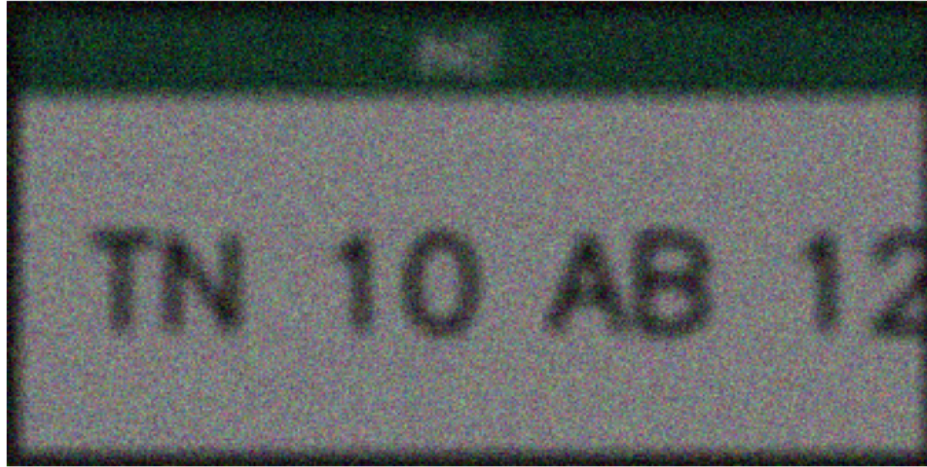
print("\nApplication:")
print("Autonomous vehicles use these techniques")
print("to improve lane detection, traffic sign")
print("recognition, and pedestrian detection.")
```

Choose Files number\_plate.jpg

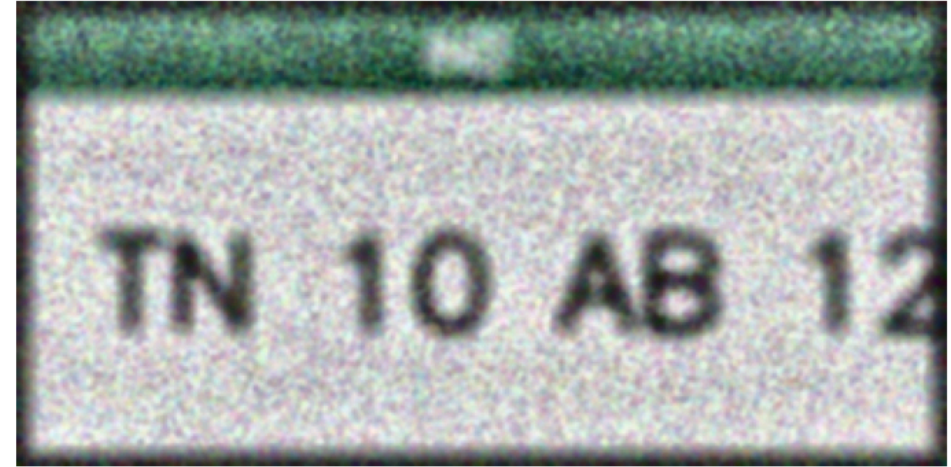
number\_plate.jpg(image/jpeg) - 14891 bytes, last modified: 7/13/2026 - 100% done

Saving number\_plate.jpg to number\_plate.jpg

Poor Image (Blur + Noise + Low Light)



Enhanced Image



#### Image Formation Model Analysis

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Original image is degraded due to:

- Blur
- Low illumination
- Sensor noise

Enhancement Techniques Used:

- Brightness correction
- Noise reduction
- Contrast enhancement (CLAHE)

Application:

Autonomous vehicles use these techniques to improve lane detection, traffic sign recognition, and pedestrian detection.

